

1

$$a) F(A, B, C, D) = \overline{(\overline{A}B + \overline{B}C)} + \overline{(D + \overline{B}C)}$$

$$= \overline{(\overline{A}B + \overline{B}C)} \cdot \overline{(D + \overline{B}C)}$$

$$= (\overline{\overline{A}B} \cdot \overline{\overline{B}C}) \cdot (D + \overline{B}C)$$

$$[x'' = x]$$

$$= [(\overline{\overline{A}} + \overline{\overline{B}}) \cdot (\overline{\overline{B}} + \overline{\overline{C}})] \cdot (D + \overline{B}C)$$

$$= (A + \overline{B}) \cdot (B + \overline{C}) \cdot (D + \overline{B}C)$$

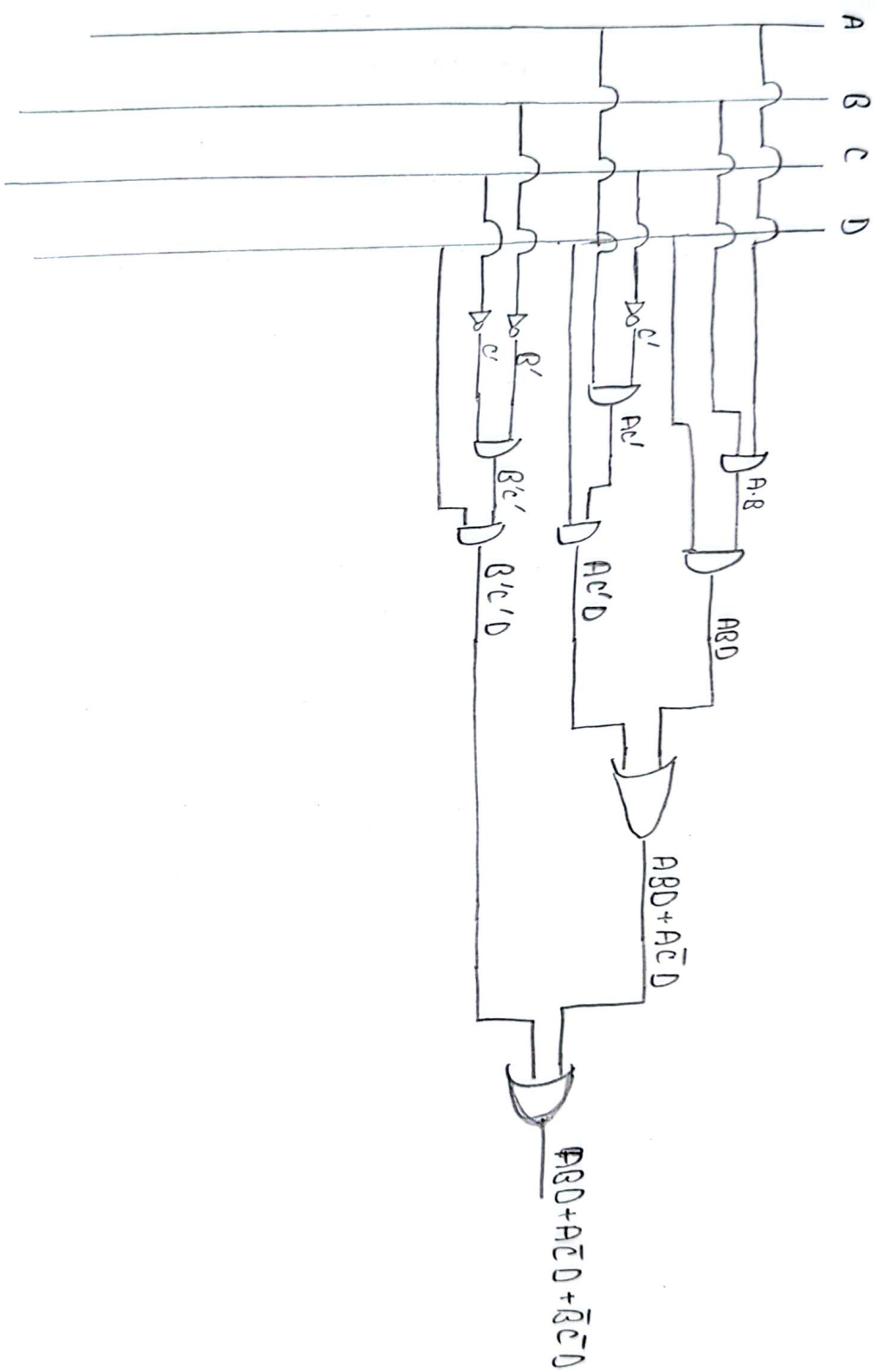
$$= (AB + A\overline{C} + B\overline{B} + \overline{B}C) (D + \overline{B}C)$$

$$[x \cdot x' = 0]$$

$$= (AB + A\overline{C} + \overline{B}C) (D + \overline{B}C)$$

$$= (ABD + A\overline{C}D + \overline{B}C\overline{D} + AB\overline{B}C + A\overline{C}\overline{B}C + \overline{B}C\overline{B}C) \quad [x \cdot x' = 0]$$

$$= ABD + A\overline{C}D + \overline{B}C\overline{D}$$



b)

Decimal	x	y	z	Output
0	0	0	0	1
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	0
7	1	1	1	0

Here, $F(x, y, z) = \pi(1, 4, 6, 7)$

$$= (x + y + z') (x' + y + z) (x' + y' + z) (x' + y' + z')$$

$$= (xx' + xy + xz + x'y + yy + yz + x'z' + yz' + zz') (x' + y + z) (x' + y' + z')$$

$$[x \cdot x' = 0]$$

$$= (xy + xz + x'y + y + yz + x'z' + yz') (x' + y + z) (x' + y' + z')$$

$$[x + x' = 1]$$

$$[1 + x = 1]$$

$$[x + x = x]$$

$$= [(y(x+x') + y(z+z') + y) + xz + x'z'] (x' + y + z) (x' + y' + z')$$

$$= [(y + y + y) + xz + x'z'] (x' + y + z) (x' + y' + z')$$

$$= (y + xz + x'z') (x' + y + z) (x' + y' + z')$$

$$= (x'y + yy' + yz + xx'z + xy'z + xzz + x'x'z' + x'y'z' + x'zz') (x' + y' + z')$$

$$= (x'y + 0 + yz + 0 + xy'z + xz + x'z' + x'y'z' + 0) (x' + y' + z')$$

$$= (x'x'y + x'y'z + xx'y'z + xx'z + x'x'z' + x'x'y'z') + (x'y'y' + yy'z + xy'y'z' + xy'z + x'y'z' + x'y'y'z') + (x'y'z' + yz'z + xy'zz' + xzz' + x'z'z' + x'y'z'z')$$

$$= x'y + x'y'z + 0 + 0 + x'z' + x'y'z' + 0 + 0 + xy'z + xy'z + x'y'z' + x'y'z' + x'y'z' + 0 + 0 + 0 + x'z' + x'y'z'$$

$$= x'y + x'y'z + x'z' + x'y'z' + xy'z + x'y'z'z' + x'y'z'$$

$$= x' (y + y'z + z' + y'z' + yz') + xy'z$$

$$= x' [y + y(z+z')] + z' + y'z'] + xy'z$$

$$[x + x' = 1]$$

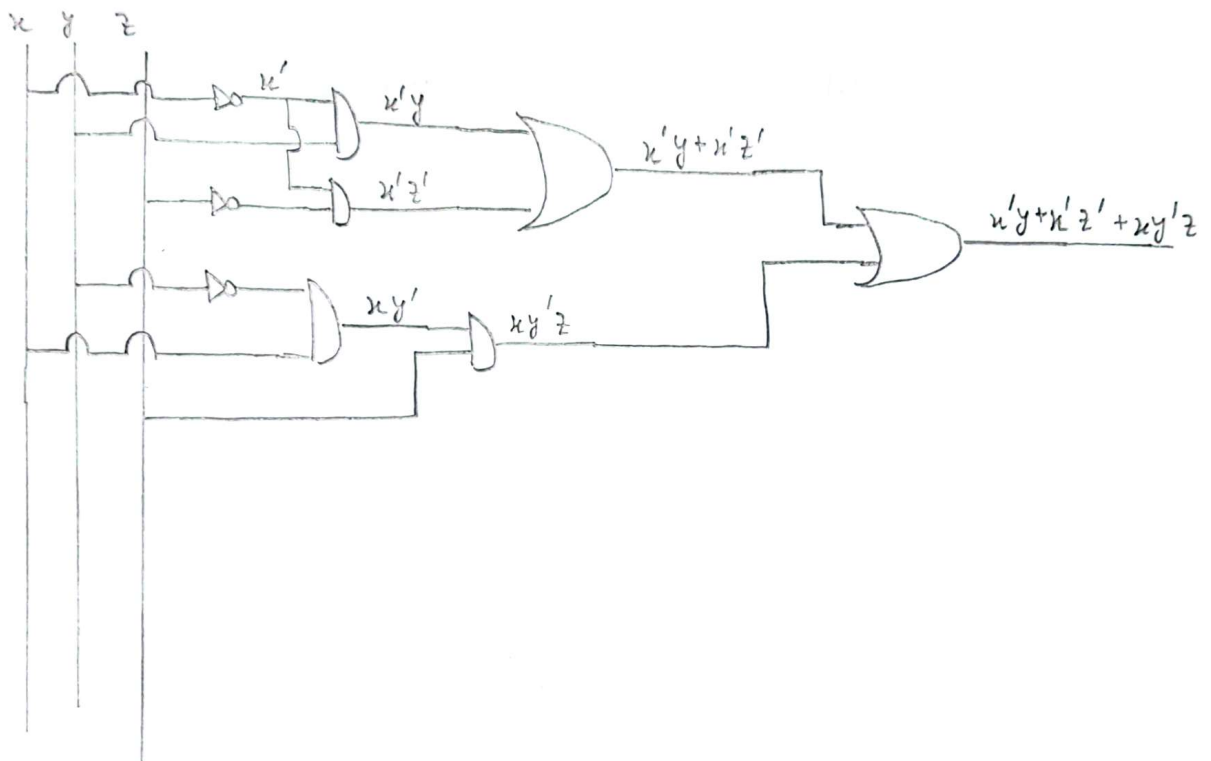
$$= x' [y + y + z'(y' + 1)] + xy'z$$

$$[x + x = x]$$

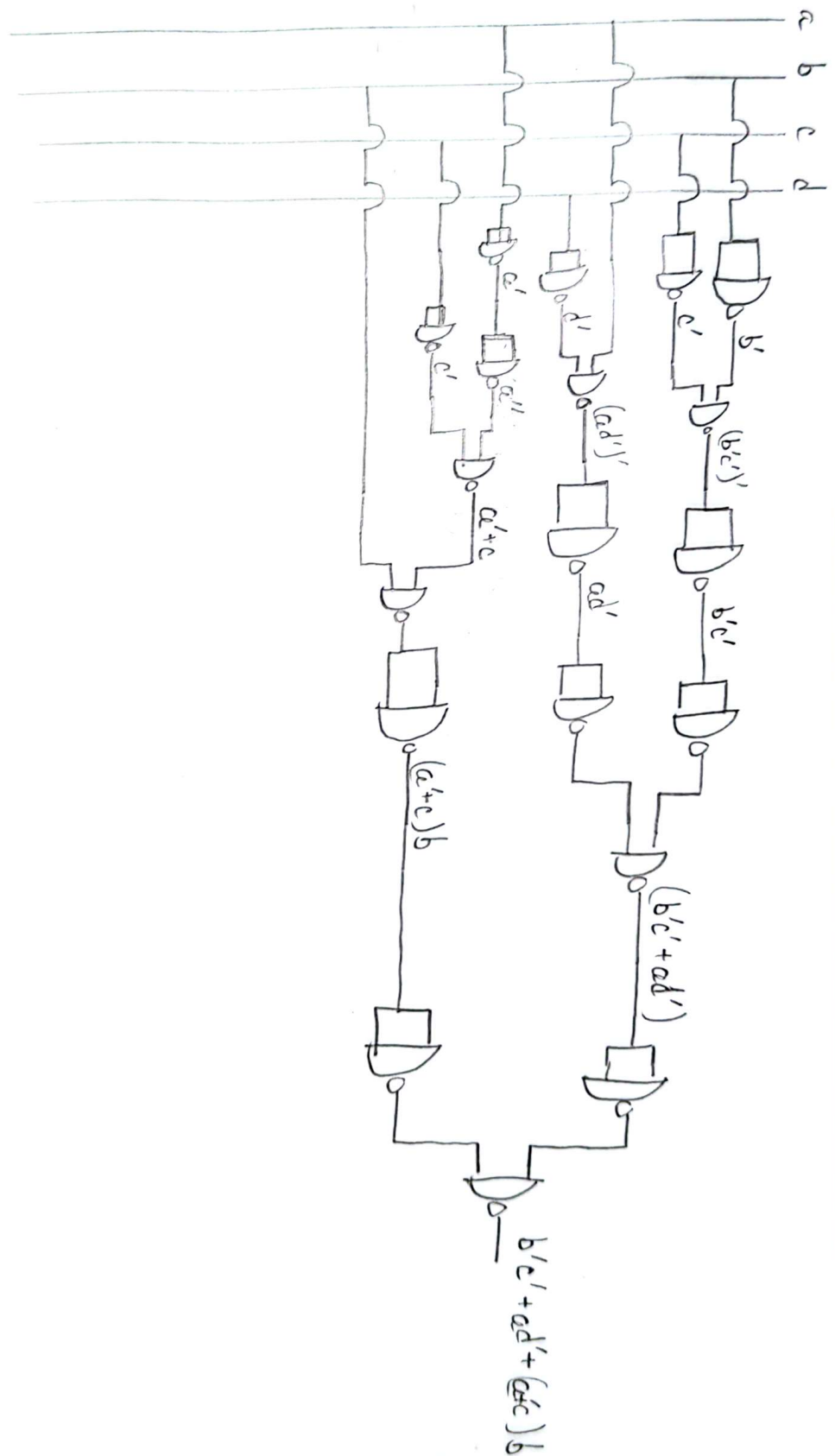
$$[x + 1 = 1]$$

$$= x' [y + z'] + xy'z$$

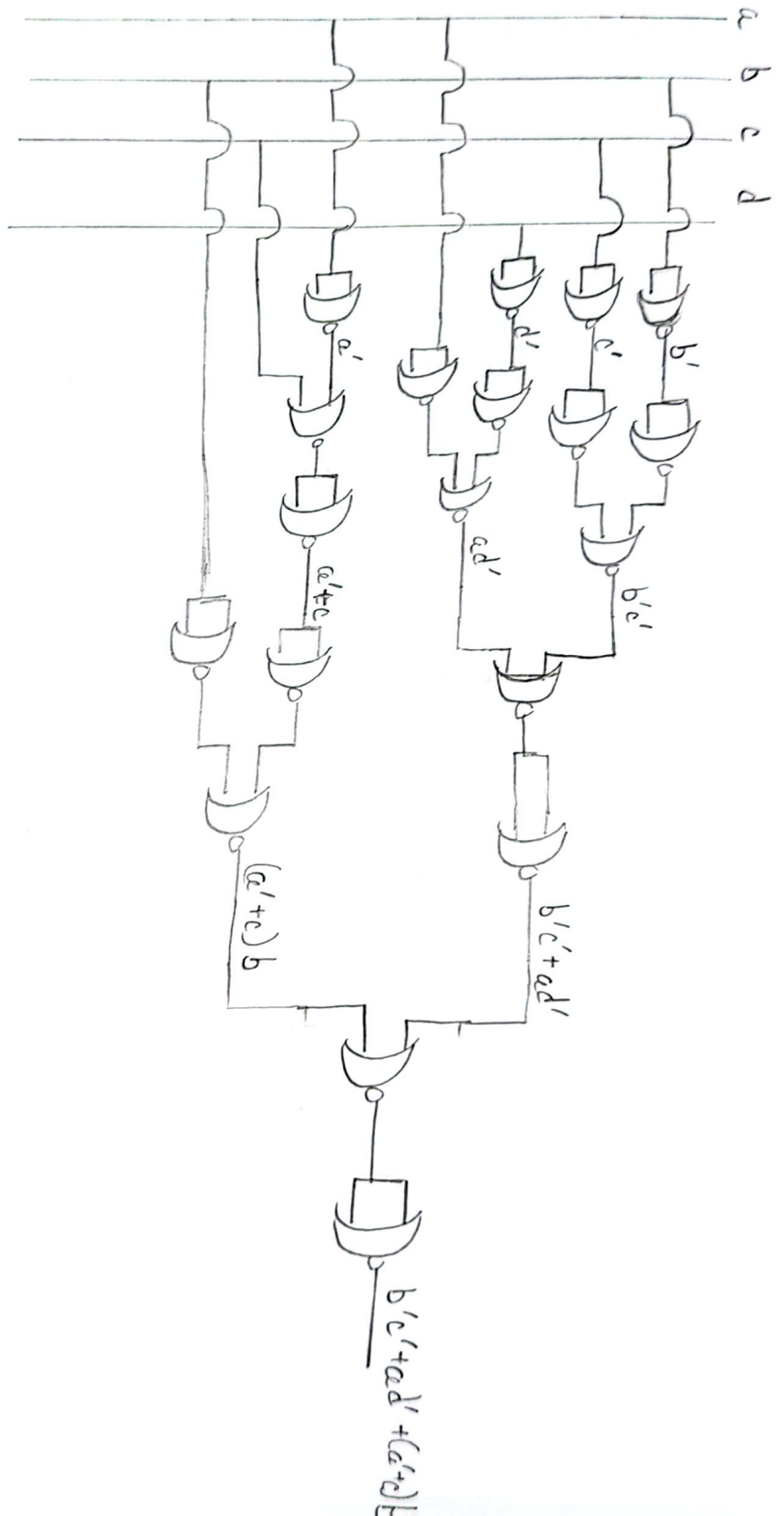
$$= x'y + x'z' + xy'z$$



2 b) Using NAND gate - $F(a,b,c,d) = b'c' + ad' + (a'+c)b$



Using NOR Gate



3

$$b) F(\omega, \kappa, \gamma, z) = \kappa' \gamma + \gamma (z + \kappa' z)$$

$$= \kappa' \gamma + \gamma z + \kappa' \gamma z$$

$$= (\kappa' \gamma + \gamma) (\kappa' \gamma + z) + \kappa' \gamma z \quad [\kappa' \gamma z = (\kappa' + \gamma) (\kappa' z)]$$

$$= \gamma (\kappa' + 1) (\kappa' \gamma + z + \kappa') (\kappa' \gamma + \gamma z)$$

$$= \gamma [\kappa' (\gamma + 1) z] (\kappa' \gamma + \gamma z)$$

$$= \gamma (\kappa' + z) (\kappa' \gamma + \gamma z)$$

$$= (\kappa' \gamma + \gamma z) (\kappa' \gamma + \gamma z)$$

$$= (\kappa' \gamma + \gamma z)$$

$$= (\kappa' \gamma + \gamma) (\kappa' \gamma + z)$$

$$= \gamma (\kappa' + 1) (z + \kappa') (z + \gamma)$$

$$= (z + \kappa') \gamma \quad [\gamma z + \gamma = \gamma]$$

$$= (\kappa' z + \gamma \gamma') (\gamma + \kappa \kappa')$$

$$= (\kappa + \gamma + z) (\kappa' + \gamma' + z) (\kappa + \gamma) (\kappa' + \gamma)$$

$$= (\kappa' + \omega \omega' + \gamma + z) (\kappa' + \omega \omega' + \gamma' + z) (\kappa + \gamma + z z') (\kappa' + \gamma + z z')$$

$$= (x' + \omega + y + z)(\omega' + x' + y + z)(\omega + x' + y' + z)(\omega' + x' + y' + z)$$

Now,

$$(x + y + z)(x + y + z')(x + y + z)(x' + y + z')$$

$$= (\omega + x + y + z)(\omega' + x + y + z)(\omega + x' + y' + z)(\omega' + x' + y' + z)(x + y + z + \omega\omega')$$

$$(x + y + z' + \omega\omega')(x' + y + z + \omega\omega')(x' + y + z' + \omega\omega')$$

$$= (\omega + x' + y + z)(\omega' + x' + y + z)(\omega + x + y' + z)(\omega' + x' + y' + z)(x + y + z + \omega)$$

$$(x + y + z + \omega')(x + y + z' + \omega)(x + y + z' + \omega')(x' + y + z + \omega)(x' + y + z + \omega')$$

$$(x + y + z' + \omega)(x' + y + z' + \omega)$$

$$= (\omega + x' + y + z)(\omega' + x' + y + z)(\omega + x' + y' + z)(\omega' + x' + y' + z)(\omega + x + y + z)$$

$$(\omega' + x + y + z)(\omega + x + y + z)(\omega' + x + y + z')$$

$$[x \cdot x = x]$$

$$= (0100, 1100, 0110, 1110, 0000, 1000, 0001, 1001)$$

$$= \pi(4, 12, 6, 14, 0, 8, 1, 9)$$

$$= \pi(0, 1, 4, 6, 8, 9, 12, 14)$$

4

a) Without De-Morgan's Law,

$$(a'bc' + a'b')$$

$$= (a' + b + c') (a' + b')$$

[Dual form]

$$= (a'' + b' + c'') \cdot (a'' + b'')$$

$$= (a + b' + c) (a + b)$$

[Complement form]

With De-Morgan's Law,

$$(a'bc' + a'b')'$$

$$= (a'bc')' (a'b')'$$

$$= [(a'b)' + c''] (a'' + b'')$$

$$= (a'' + b' + c'') (a'' + b'')$$

$$= (a + b' + c) (a + b)$$

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$$b) F(P, Q, R, S) = Q' + RS' + QS$$

$$= Q'(P + P') + RS(P + P') + QS(P + P')$$

$$= PQ' + PQ' + RS'P + RS'P' + PQS + P'QS$$

$$= PQ'(R + R') + PQ'(R + R') + PRS'(Q + Q') + P'S'R(Q + Q') + PQS(R + R') + P'QS(R + R')$$

$$= PQ'R + PQ'R' + PQ'R + PQ'R' + PQRS' + PQ'R S' + P'QS'R + P'Q'S'R + PQRS + PQR'S + P'QRS + P'QR'S$$

$$= PQ'R(S + S') + PQ'R'(S + S') + PQ'R(S + S') + PQ'R'(S + S') + PQRS' + PQ'R S' + P'QRS' + P'Q'S'R + PQRS + PQR'S + P'QRS + P'QR'S$$

$$= PQ'RS + PQ'R S' + PQ'R'S + PQ'R'S' + PQ'RS + PQ'R S' + PQ'R'S + PQ'R'S' + P'QRS' + P'Q'S'R + PQRS + PQR'S + P'QRS + P'QR'S$$

$$[X + X = X]$$

$$= PQ'RS + PQ'RS' + PQ'R'S + PQ'R'S' + P'Q'RS + P'Q'RS' +$$

$$P'Q'R'S + PQRS' + P'QRS' + PQRS + PQR'S + PQ'R'S + P'QPS$$

[Canonical SOP]

$$= (1011, 1010, 1001, 1000, 0011, 0010, 0001, 1110, 0110, 1111, 1101, 0111, 0101)$$

$$= \Sigma (11, 10, 9, 8, 3, 2, 1, 14, 6, 15, 13, 7, 5)$$

$$= \Sigma (1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 13, 14, 15) \quad [\text{Minterm}]$$

$$\therefore \text{Maxterm} = \Pi (0, 4, 12)$$

$$\therefore \text{Canonical POS} = (0000, 0100, 1100)$$

$$= (PQRS, PQ'RS, P'Q'RS)$$